Northeastern University
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A project report on

Public Health Ontario System

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Table of Contents

Sr. No	Title	Page No
1.	Purpose of the Project	3
2.	Project Outline	4
3.	Project Outcomes/Key Deliverables	6
4.	Business Case	7
5.	Tools and Techniques	9
6.	Conceptual Data Model	10
7.	Initial ERD	12
8.	Findings and Enhancements in Initial ERD	13
9.	Final ERD	14
10.	Business Rule	15
11.	DDL Scripts	17
12.	DML Scripts	25
13.	DCL Scripts	92
14.	Queries	95
15.	Challenges	97
16.	Conclusion	98

1. Purpose of the Project

The purpose of the Public Health Ontario Systems Database project is to establish a comprehensive and integrated information management system that enhances the efficiency, accuracy, and accessibility of healthcare data across the public health domain. By creating a centralized database, the project aims to streamline critical aspects of healthcare management, including patient records, appointment scheduling, prescription management, and healthcare provider coordination. The overarching goal is to improve patient care outcomes by providing healthcare professionals with a holistic view of patient history, facilitating informed decision-making, and promoting seamless collaboration among different healthcare departments and providers.

Furthermore, the project seeks to enhance the overall effectiveness of public health initiatives by leveraging data for surveillance, research, and reporting purposes. By consolidating information on disease outbreaks, health trends, and treatment outcomes, the database enables public health officials to proactively respond to emerging health challenges, implement preventive measures, and make data-driven policy decisions. The project's purpose extends beyond immediate healthcare delivery to contribute to the broader goals of advancing public health research, supporting evidence-based practices, and ultimately improving the overall health and well-being of the community served by Public Health Ontario.

2. Project Outline

Overview of the Project The Public Health Ontario System (PHO System) is a comprehensive, centralized that integrates various components of the public health sector in Ontario. The primary goal of this project is to improve the efficiency and effectiveness of the Public Health System, leading to better healthcare services and outcomes for the people of Ontario.

Objectives of the Project

- To enhance the user-friendliness and overall performance of the Public Health System.
- To implement robust data management capabilities, ensuring accurate and up-to-date information is available.
- To optimize the overall processes and procedures within the Healthcare System, leading to streamlined workflows and improved efficiency.

Problems with the Existing System

- Insufficient data integration capabilities, leading to redundancies and duplications in stored information.
- Slow data retrieval times, causing delays in providing timely healthcare services to the public.
- Outdated and non-user-friendly software interface, resulting in lower operational efficiency and reduced adoption of the system by healthcare professionals.

Solutions Proposed

- Utilize advanced technologies and tools, such as Oracle Sql Developer.
- Implement a comprehensive data management system that effectively manages, integrates, and distributes all relevant healthcare data.

Expected Outcomes

- A highly efficient, user-friendly, and robust public health information system that provides accurate, timely, and easily accessible healthcare information to all stakeholders in the Ontario public health sector.
- Enhanced patient care and outcomes through the optimized use of the database.

Challenges and Limitations

- Resistance to change from healthcare professionals who may be unwilling to adopt the new system.
- Potential data privacy and security concerns that may arise from the integration of various data sources within the Healthcare database.

- Budgetary constraints that may impact the ability to implement all proposed enhancements and upgrades.

Conclusion The Public Health Ontario System database is an essential tool. By addressing the challenges and limitations associated with the existing system, this project has the potential to revolutionize the way public health information is managed and utilized in Ontario, ultimately leading to improved healthcare services and outcomes for the people of this province.

3. Project Outcomes/Key Deliverables

The project outcomes and key deliverables for the Public Health Ontario Systems Database project can be summarized as follows:

I. **Project Overview and Objectives:**

- Clearly defined project goals and objectives, emphasizing the need for an integrated database system in the public health context.

II. **Stakeholder Analysis:**

- Identification and analysis of key stakeholders, including healthcare providers, administrators, patients, and public health officials.

III. **Requirements Analysis:**

- Detailed documentation of user requirements, data requirements gathered through group discussion.

IV. **Database Design:**

- Presentation of the database design, including the Entity-Relationship Diagram (ERD), and normalized database schema.

V. **Project Timeline and Milestones:**

- Presentation of the project timeline, highlighting major milestones achieved and providing a visual representation of the project's progress.

VI. **Lessons Learned:**

- Reflection on challenges faced, successes achieved, and lessons learned throughout the project. This section aims to inform future projects and improve the overall project management process.

VII. **Conclusion and Recommendations:**

- A concluding section summarizing the project outcomes, key deliverables, and providing recommendations for ongoing improvements or enhancements.

4. Business Case

1. **Executive Summary:** The Public Health Ontario Systems Database Implementation project addresses critical challenges within the public health domain, aiming to enhance the efficiency, accuracy, and accessibility of healthcare data. The project envisions the creation of a centralized information management system to optimize patient care, streamline healthcare provider coordination, and support public health initiatives. This business case outlines the rationale, benefits, and strategic importance of implementing the database system.

2. **Problem Statement:** Current healthcare systems face challenges related to fragmented data, inefficient information flow, and limited coordination among healthcare providers. These issues hinder the delivery of seamless and integrated healthcare services, impacting patient outcomes and public health management.

3. **Project Objectives:**

- Improve Patient Care: Provide healthcare professionals with a comprehensive view of patient information to support informed decision-making and personalized care.
- Enhance Provider Coordination: Facilitate seamless communication and collaboration among healthcare providers, departments, and integrated care teams.
- Support Public Health Initiatives: Contribute to public health surveillance, research, and reporting efforts by consolidating and analyzing health data.

4. **Strategic Alignment:** The project aligns with the strategic goals of improving healthcare delivery, promoting patient safety, and advancing public health research within the province of Ontario. It directly supports the mission of Public Health Ontario to protect and improve the health of the population.

5. **Benefits:**

- Improved Patient Outcomes: Timely access to accurate patient information leads to more informed diagnoses, personalized treatment plans, and improved overall healthcare outcomes.
- Enhanced Efficiency: Streamlined processes, appointment scheduling, and integrated care departments result in more efficient use of healthcare resources.
- Public Health Insights: The database supports data-driven decision-making for public health officials, helping identify trends, monitor disease outbreaks, and implement preventive measures.

6. **Project Costs:** While there will be initial costs associated with system development, deployment, and training, the long-term benefits in terms of improved healthcare delivery, reduced operational inefficiencies, and better public health management justify the investment.

7. **Risks and Mitigation:** Potential risks include data security concerns, system integration challenges, and resistance to change from healthcare professionals. Mitigation strategies include robust data security measures, phased implementation, and comprehensive training programs.

8. **Return on Investment (ROI):** The expected ROI includes improved healthcare outcomes, increased operational efficiency, and enhanced public health management capabilities. The reduction in manual processes and improved data accuracy will contribute to long-term cost savings.

This business case serves as a foundation for decision-makers, providing a comprehensive understanding of the project's strategic importance, anticipated benefits, and alignment with organizational goals. It supports the justification for investing resources in the implementation of the database system.

5. Tools and Techniques

Tools and Techniques for Public Health Ontario Systems Database Implementation Project:

1. **Database Design and Modeling:**

- Tools: Oracle SQL Developer, draw.io
- Techniques:Database and Tables creation, rows insertion, Entity-Relationship Diagrams (ERD)
- Purpose: Designing and visualizing the database structure, relationships, and data flow.

2. **Communication and Collaboration:**

- Tools: Microsoft Teams, Whatsapp
- Techniques: Regular status meetings
- Purpose: Facilitating communication and collaboration among project team members

3. **Training and Documentation:**

- Tools: Google Docs, Google Slides
- Techniques: Project Report, Project Presentation
- Purpose: Providing comprehensive documentation and presentation.

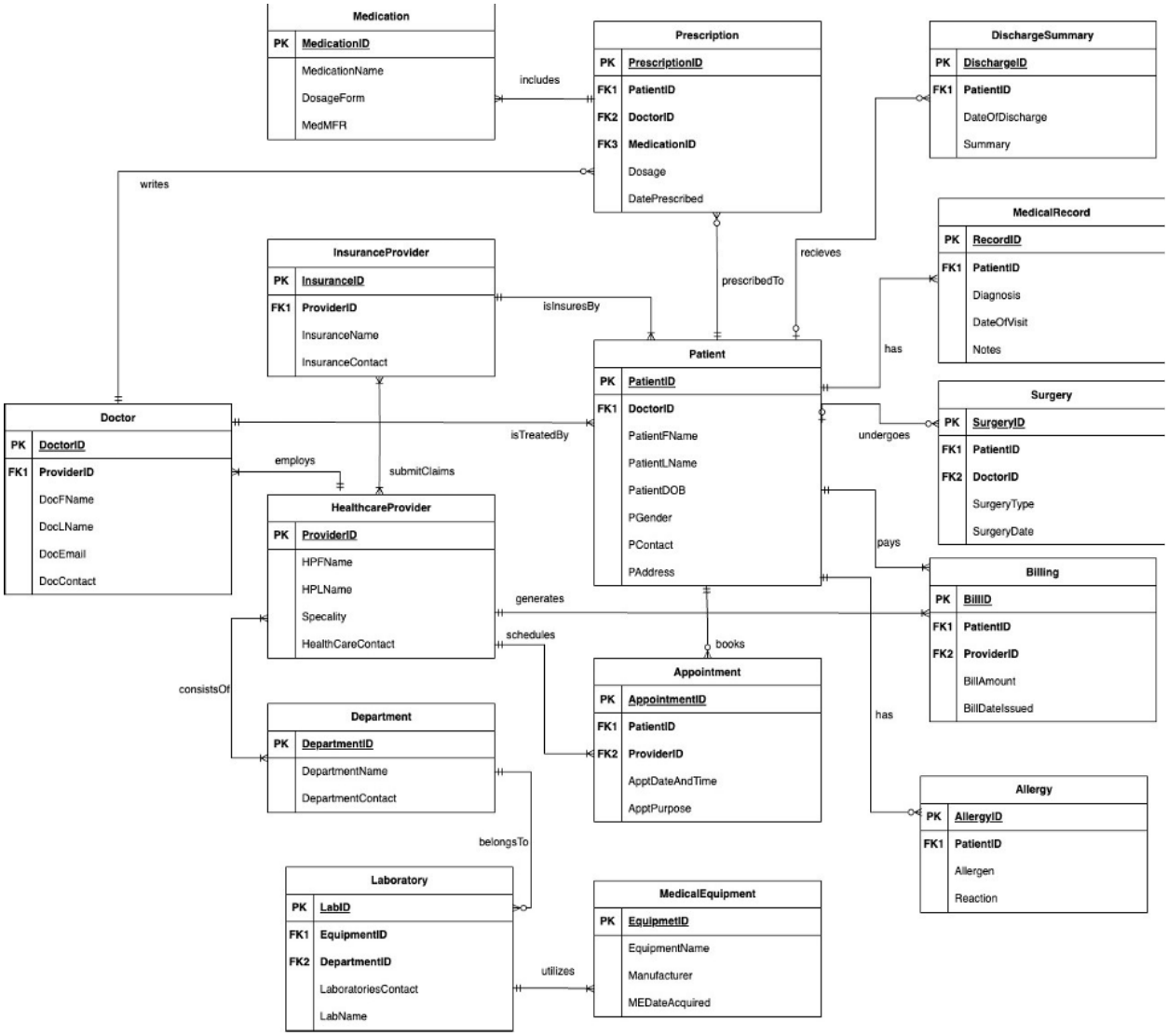
6. Conceptual Data Model

Entities and Attributes

Sr No.	Entity	Attribute
1.	**Patients**	PatientID (Primary Key), DoctorID (Foreign Key), PatientFName, PatientLName, PatientDOB, PGender, PContact, PAddress
2.	**MedicalRecords**	RecordID (Primary Key), PatientID (Foreign Key), Diagnosis, DateOfVisit, Notes
3.	**Appointments**	AppointmentID (Primary Key), PatientID (Foreign Key), ProviderID (Foreign Key), AppDateAndTime, ApptPurpose
4.	**HealthCareProviders**	ProviderID (Primary Key), AdministrationID (Foreign Key), HPLName, Speciality, HealthCareContact
5.	**Departments**	DepartmentID (Primary Key), ICDepartmentID (Foreign Key), DepartmentName, DepartmentContact
6.	**InsuranceProviders**	InsuranceID (Primary Key), AdministrationID (Foreign Key), InsuranceName, InsuranceContact
7.	**Prescriptions**	PrescriptionID (Primary Key), PatientID (Foreign Key), DoctorID (Foreign Key), MedicationID (Foreign Key), Dosage, DatePrescribed

8.	**Billing**	BillID (Primary Key), PatientID (Foreign Key), ProviderID (Foreign Key), BillAmount, BillDateIssued
9.	**Laboratories**	LabID (Primary Key), EquipmentID (Foreign Key), DepartmentID (Foreign Key), LaboratoriesContact, LabName
10.	**ImagingCenters**	ImagingCenterID (Primary Key), EquipmentID (Foreign Key), DepartmentID (Foreign Key), ImagingCenterContact, ImagingCenterName
11.	**Medications**	MedicationID (Primary Key), MedicationName, DosageForm, MedMFR
12.	**Allergies**	AllergyID (Primary Key), PatientID (Foreign Key), Allergen, Reaction
13.	**Scheduling**	ScheduleID (Primary Key), ProviderID (Foreign Key), AppointmentID (Foreign Key), ScheduleDate, ScheduleTime
14.	**Surgeries**	SurgeryID (Primary Key), PatientID (Foreign Key), DoctorID (Foreign Key), SurgeryType, SurgeryDate
15.	**MedicalEquipment**	EquipmentID (Primary Key), EquipmentName, Manufacturer, MedDateAcquired
16.	**DischargeSummaries**	DischargeID (Primary Key), PatientID (Foreign Key), DateOfDischarge, Summary
17.	**IntegratedCareDepartment**	CDepartmentID (Primary Key), ProviderID (Foreign Key), DepartmentContact

7. Initial ERD



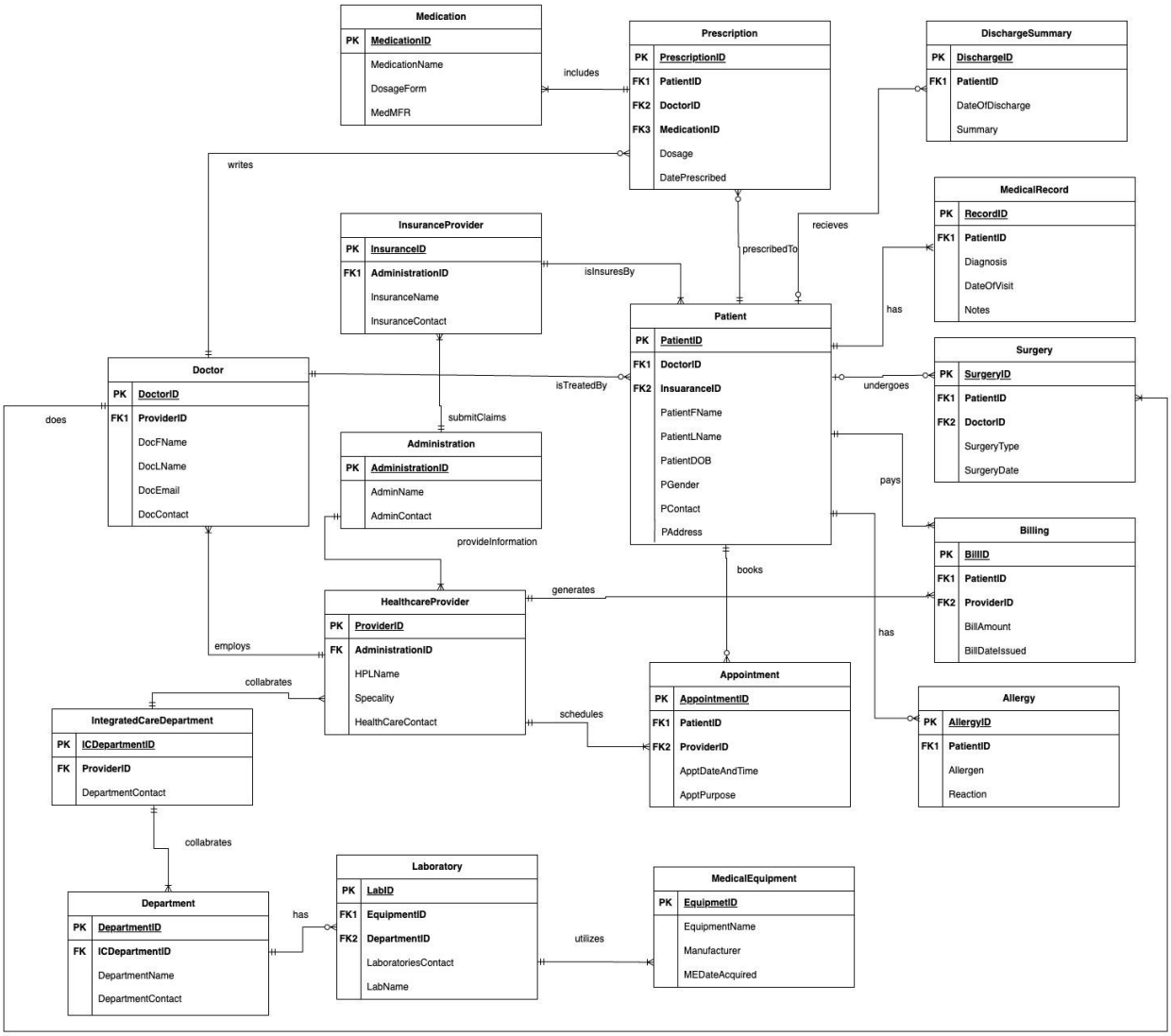
8. Findings and Enhancements in Initial ERD

The initial Entity-Relationship Diagram (ERD) for the Public Health Ontario Systems Database was submitted for review, marked as a pivotal milestone in the project's development.

Following a thorough evaluation by the Teaching Assistant (TA), Harshit Prajapati, constructive feedback was provided, emphasizing the need to remove many-to-many relationships to enhance the database's normalization. This crucial feedback underscores the importance of achieving a well-structured and efficient database design.

In response to the TA's guidance, a strategic decision was made to replace many-to-many relationships with either zero-to-many (0-N) or one-to-many (1-N) relationships. This choice is driven by the principles of normalization, aiming to reduce data redundancy, improve data integrity, and streamline the database structure. The process of refining relationships to either 0-N or 1-N signifies a commitment to adhering to best practices in database design and ensures that the resulting schema aligns with the fundamental principles of relational databases. This iterative refinement, guided by valuable feedback, not only contributes to the optimization of the database but also reflects a dedication to delivering a robust and well-architected system for the Public Health Ontario project.

9. Final ERD


Medication
PK MedicationID
MedicationName
DosageForm
MedMFR
includes
Prescription
PK PrescriptionID
FK1 PatientID
FK2 DoctorID
FK3 MedicationID
Dosage
DatePrescribed
DischargeSummary
PK DischargeID
FK1 PatientID
DateOfDischarge
Summary
writes
recieves
prescribedTo
MedicalRecord
PK RecordID
FK1 PatientID
Diagnosis
DateOfVisit
Notes
InsuranceProvider
PK InsuranceID
FK1 AdministrationID
InsuranceName
InsuranceContact
isInsuresBy
has
Patient
PK PatientID
FK1 DoctorID
FK2 InsuaranceID
PatientFName
PatientLName
PatientDOB
PGender
PContact
PAddress
Doctor
PK DoctorID
FK1 ProviderID
DocFName
DocLName
DocEmail
DocContact
isTreatedBy
undergoes
Surgery
PK SurgeryID
FK1 PatientID
FK2 DoctorID
SurgeryType
SurgeryDate
does
submitClaims
Administration
PK AdministrationID
AdminName
AdminContact
pays
Billing
PK BillID
FK1 PatientID
FK2 ProviderID
BillAmount
BillDateIssued
provideInformation
books
generates
HealthcareProvider
PK ProviderID
FK AdministrationID
HPLName
Speciality
HealthCareContact
has
employs
collabrates
Appointment
PK AppointmentID
FK1 PatientID
FK2 ProviderID
ApptDateAndTime
ApptPurpose
Allergy
PK AllergyID
FK1 PatientID
Allergen
Reaction
schedules
IntegratedCareDepartment
PK ICDepartmentID
FK ProviderID
DepartmentContact
collabrates
Laboratory
PK LabID
FK1 EquipmentID
FK2 DepartmentID
LaboratoriesContact
LabName
MedicalEquipment
PK EquipmetID
EquipmentName
Manufacturer
MEDateAcquired
Department
PK DepartmentID
FK ICDepartmentID
DepartmentName
DepartmentContact
has
utilizes

10. Business Rule

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1. Patient Information Confidentiality:
Rule: All patient information stored in the database is strictly confidential and must adhere to privacy regulations (e.g., Health Information Protection Act). Access to patient data is restricted to authorized healthcare providers and personnel for legitimate healthcare purposes.

2. Data Accuracy and Integrity:
Rule: The database must maintain accurate and up-to-date information. All data entries, updates, and modifications must be validated and adhere to predefined data integrity standards to ensure the reliability of healthcare records.

3. Prescription Authorization:
Rule: Only licensed healthcare providers, specifically authorized to prescribe medications, are allowed to enter and modify prescription information in the database. Prescription details must include the medication name, dosage, and prescribing date.

4. Appointment Scheduling Constraints:
Rule: Appointment scheduling must adhere to defined constraints, such as healthcare provider availability, appointment duration, and maximum daily appointments per provider. The system should automatically validate and enforce these constraints during scheduling.

5. Billing and Insurance Compliance:
Rule: Billing entries must comply with the established fee schedules and insurance policies. The system should validate billing information for accuracy, and any discrepancies should be flagged for review by authorized personnel.

6. Allergy Notification:
Rule: Healthcare providers must be alerted to patient allergies when prescribing medications or scheduling medical procedures. The system should display allergy information prominently during relevant interactions to prevent adverse reactions.

7. Laboratory and Imaging Result Confidentiality:
Rule: Laboratory and imaging results are confidential and should only be accessible to authorized healthcare providers involved in the patient's care. Access to these results is restricted, and any sharing must comply with privacy regulations.

8. Data Backup and Recovery:
Rule: Regular database backups must be conducted to ensure data integrity and facilitate recovery in case of system failures or data loss. Backup procedures should be well-documented, and recovery tests should be performed periodically.

9. User Authentication and Authorization:
Rule: Access to the database is restricted based on user roles and responsibilities. User authentication is required, and access permissions are granted based on the principle of least privilege to ensure security and prevent unauthorized access.

10. Change Management:
Rule: Any changes to the database schema, system configurations, or business rules must follow a formal change management process. Changes should be documented, reviewed, and approved before implementation to prevent unintended consequences.

11. Reporting Accuracy:
Rule: Reports generated from the database must accurately reflect the current state of healthcare data. Regular audits and reconciliation processes should be in place to ensure the accuracy and reliability of reports generated for decision-making.

12. Emergency Access Protocol:
Rule: In case of emergencies, designated personnel may be granted temporary access to patient information for the purpose of providing urgent healthcare services. All emergency access instances must be documented and reviewed for compliance.

These business rules serve as foundational principles guiding the design, implementation, and operation of the Public Health Ontario Systems Database. They contribute to data security, accuracy, and regulatory compliance, ensuring a robust and ethical healthcare information management system.

Consistent Data Types: We have ensured that data types and sizes are consistent across tables and meet the requirements of the application.

These business rules help define the structure and relationships within the Oracle SQL database project, ensuring accurate and meaningful data storage for patient and appointment information. Adjustments may be necessary based on additional project requirements or specific business needs.

11. DDL Scripts

In the Public Health Ontario Systems Database project, the implementation of Data Definition Language (DDL) queries is fundamental for shaping the structure and organization of the database. DDL queries are instrumental in defining, altering, and managing the database schema, including the creation and modification of tables, constraints, and relationships. When initiating the project, DDL CREATE TABLE queries are employed to define the structure of entities like Patients, MedicalRecords, Appointments, and others, specifying attributes, primary keys, and foreign keys. As the project evolves, DDL ALTER TABLE queries are crucial for making structural adjustments, whether it be adding new attributes or modifying existing ones to accommodate evolving data requirements. The establishment of referential integrity and constraints, such as foreign key constraints linking different tables, is achieved through DDL queries, ensuring data consistency and accuracy.
Furthermore, DDL queries are pivotal for creating indexes on specific columns, optimizing data retrieval performance. The utilization of DDL queries in this project is strategic, contributing to the creation, modification, and maintenance of a robust and well-organized database structure that aligns with the evolving needs of the Public Health Ontario System.

We have attached the script below which includes all the DDL queries performed as part of the project.

DDL Scripts Queries

```
----START------
CREATE TABLE medication (
   medicationid   INT PRIMARY KEY,
   medicationname VARCHAR(255),
   dosageform     VARCHAR(100),
   medmfr         VARCHAR(255)
);
CREATE TABLE administration (
   administrationid INT PRIMARY KEY,
   adminname        VARCHAR(255),
   admincontact     VARCHAR(20)
);
CREATE TABLE medicalequipment (
   equipmentid     INT PRIMARY KEY,
   equipmentname   VARCHAR(255),
   manufacturer    VARCHAR(255),
   meddateacquired DATE
);
CREATE TABLE healthcareprovider (
   providerid        INT PRIMARY KEY,
   administrationid  INT,
   hplname           VARCHAR(255),
   speciality        VARCHAR(100),
   healthcarecontact VARCHAR(20),
   FOREIGN KEY ( administrationid )
      REFERENCES administration ( administrationid )
);
CREATE TABLE doctor (
   doctorid   INT PRIMARY KEY,
   providerid INT,
   docfname   VARCHAR(255),
   doclname   VARCHAR(255),
   docemail   VARCHAR(100),
   doccontact VARCHAR(20),
   FOREIGN KEY ( providerid )
      REFERENCES healthcareprovider ( providerid )
```

```
);
CREATE TABLE insuranceprovider (
   insuranceid      INT PRIMARY KEY,
   administrationid INT,
   insurancename    VARCHAR(255),
   insurancecontact VARCHAR(20),
   FOREIGN KEY ( administrationid )
      REFERENCES administration ( administrationid )
);
CREATE TABLE patient (
   patientid    INT PRIMARY KEY,
   doctorid     INT,
   patientfname VARCHAR(255),
   patientlname VARCHAR(255),
   patientdob   DATE,
   pgender      VARCHAR(1),
   pcontact     VARCHAR(20),
   paddress     VARCHAR(255),
   insuranceid  INT,
   FOREIGN KEY ( insuranceid )
      REFERENCES insuranceprovider ( insuranceid ),
   FOREIGN KEY ( doctorid )
      REFERENCES doctor ( doctorid )
);
CREATE TABLE integratedcaredepartment (
   icdepartmentid    INT PRIMARY KEY,
   providerid        INT,
   departmentcontact VARCHAR(20),
   FOREIGN KEY ( providerid )
      REFERENCES healthcareprovider ( providerid )
);
CREATE TABLE department (
   departmentid      INT PRIMARY KEY,
   icdepartmentid    INT,
   departmentname    VARCHAR(255),
   departmentcontact VARCHAR(20),
   FOREIGN KEY ( icdepartmentid )
      REFERENCES integratedcaredepartment ( icdepartmentid )
);
CREATE TABLE laboratory (
```

```
    labid              INT PRIMARY KEY,
    equipmentid        INT,
    departmentid       INT,
    laboratoriescontact VARCHAR(20),
    labname            VARCHAR(255),
    FOREIGN KEY ( equipmentid )
       REFERENCES medicalequipment ( equipmentid ),
    FOREIGN KEY ( departmentid )
       REFERENCES department ( departmentid )
);
CREATE TABLE appointment (
    appointmentid  INT PRIMARY KEY,
    patientid      INT,
    providerid     INT,
    appdateandtime DATE,
    apptpurpose    VARCHAR2(255),
    FOREIGN KEY ( patientid )
       REFERENCES patient ( patientid ),
    FOREIGN KEY ( providerid )
       REFERENCES healthcareprovider ( providerid )
);
CREATE TABLE prescription (
    prescriptionid INT PRIMARY KEY,
    patientid      INT,
    doctorid       INT,
    medicationid   INT,
    dosage         VARCHAR(50),
    dateprescribed DATE,
    FOREIGN KEY ( patientid )
       REFERENCES patient ( patientid ),
    FOREIGN KEY ( doctorid )
       REFERENCES doctor ( doctorid ),
    FOREIGN KEY ( medicationid )
       REFERENCES medication ( medicationid )
);
CREATE TABLE dischargesummary (
    dischargeid     INT PRIMARY KEY,
    patientid       INT,
    dateofdischarge DATE,
    summary         VARCHAR(250),
```

```
    FOREIGN KEY ( patientid )
        REFERENCES patient ( patientid )
);
CREATE TABLE medicalrecord (
    recordid    INT PRIMARY KEY,
    patientid   INT,
    diagnosis   VARCHAR(255),
    dateofvisit DATE,
    notes       VARCHAR(250),
    FOREIGN KEY ( patientid )
        REFERENCES patient ( patientid )
);
CREATE TABLE surgery (
    surgeryid   INT PRIMARY KEY,
    patientid   INT,
    doctorid    INT,
    surgerytype VARCHAR(255),
    surgerydate DATE,
    FOREIGN KEY ( patientid )
        REFERENCES patient ( patientid ),
    FOREIGN KEY ( doctorid )
        REFERENCES doctor ( doctorid )
);
CREATE TABLE billing (
    billid        INT PRIMARY KEY,
    patientid     INT,
    providerid    INT,
    billamount    DECIMAL(10, 2),
    billdateissued DATE,
    FOREIGN KEY ( patientid )
        REFERENCES patient ( patientid ),
    FOREIGN KEY ( providerid )
        REFERENCES healthcareprovider ( providerid )
);
CREATE TABLE allergy (
    allergyid INT PRIMARY KEY,
    patientid INT,
    allergen  VARCHAR(255),
    reaction  VARCHAR(255),
    FOREIGN KEY ( patientid )
```

```
        REFERENCES patient ( patientid )
);
-----indexing----
-- Indexing for Medication table
CREATE INDEX idx_medication_medicationid ON
    medication (
        medicationid
    );
--CREATE INDEX idx_Medication_MedicationName ON Medication(MedicationName);
-- Indexing for Administration table
CREATE INDEX idx_administration_administrationid ON
    administration (
        administrationid
    );
--CREATE INDEX idx_Administration_AdminName ON Administration(AdminName);
-- Indexing for Medicalequipment table
CREATE INDEX idx_medicalequipment_equipmentid ON
    medicalequipment (
        equipmentid
    );
--CREATE INDEX idx_Medicalequipment_EquipmentName ON
Medicalequipment(EquipmentName);
-- Indexing for Healthcareprovider table
CREATE INDEX idx_healthcareprovider_providerid ON
    healthcareprovider (
        providerid
    );
--CREATE INDEX idx_Healthcareprovider_HPLName ON Healthcareprovider(HPLName);
-- Indexing for Doctor table
CREATE INDEX idx_doctor_doctorid ON
    doctor (
        doctorid
    );
--CREATE INDEX idx_Doctor_DocFName ON Doctor(DocFName);
-- Indexing for Patient table
CREATE INDEX idx_patient_patientid ON
    patient (
        patientid
    );
--CREATE INDEX idx_Patient_PatientFName ON Patient(PatientFName);
```

```
-- Indexing for Insuranceprovider table
CREATE INDEX idx_insuranceprovider_insuranceid ON
   insuranceprovider (
      insuranceid
   );
--CREATE INDEX idx_Insuranceprovider_InsuranceName ON
Insuranceprovider(InsuranceName);
-- Indexing for Integratedcaredepartment table
CREATE INDEX idx_integratedcaredepartment_icdepartmentid ON
   integratedcaredepartment (
      icdepartmentid
   );
--CREATE INDEX idx_Integratedcaredepartment_DepartmentName ON
Integratedcaredepartment(DepartmentName);
-- Indexing for Department table
CREATE INDEX idx_department_departmentid ON
   department (
      departmentid
   );
--CREATE INDEX idx_Department_DepartmentName ON Department(DepartmentName);
-- Indexing for Laboratory table
CREATE INDEX idx_laboratory_labid ON
   laboratory (
      labid
   );
--CREATE INDEX idx_Laboratory_LabName ON Laboratory(LabName);
-- Indexing for Appointment table
CREATE INDEX idx_appointment_appointmentid ON
   appointment (
      appointmentid
   );
-- Indexing for Prescription table
CREATE INDEX idx_prescription_prescriptionid ON
   prescription (
      prescriptionid
   );
-- Indexing for Dischargesummary table
CREATE INDEX idx_dischargesummary_dischargeid ON
   dischargesummary (
      dischargeid
```

```
    );
-- Indexing for MedicalRecord table
CREATE INDEX idx_medicalrecord_recordid ON
    medicalrecord (
        recordid
    );
-- Indexing for Surgery table
CREATE INDEX idx_surgery_surgeryid ON
    surgery (
        surgeryid
    );
-- Indexing for Billing table
CREATE INDEX idx_billing_billid ON
    billing (
        billid
    );
-- Indexing for Allergy table
CREATE INDEX idx_allergy_allergyid ON
    allergy (
        allergyid
    );
----END----
```

12. DML Scripts

In the Public Health Ontario Systems Database project, the use of Data Manipulation Language (DML) queries plays a pivotal role in managing and interacting with the database's content. DML queries are instrumental in facilitating various operations, including the insertion, updating, retrieval, and deletion of data within the database. For instance, when a new patient is registered, DML INSERT queries are employed to add their information into the Patients table, capturing essential details such as patient name, contact information, and associated healthcare provider. Likewise, DML UPDATE queries enable the modification of existing records, ensuring that patient information remains accurate and up-to-date. Appointment scheduling, a critical aspect of healthcare management, relies on DML queries to INSERT new appointments into the Appointments table, linking patients with specific healthcare providers at designated dates and times.

The judicious use of DML queries is fundamental to the dynamic and responsive nature of the database, ensuring that healthcare practitioners have real-time access to critical information, ultimately contributing to improved patient care and efficient healthcare operations.

We have attached the script below which includes all the DML queries performed as part of the project.

DML Scripts Queries

```
------INSERT queries-----
---Medication---
-- Inserting record 1
INSERT INTO medication (
    medicationid,
    medicationname,
    dosageform,
    medmfr
) VALUES (
    1,
    'Aspirin',
    'Tablet',
    'Bayer'
);
-- Inserting record 2
INSERT INTO medication (
    medicationid,
    medicationname,
    dosageform,
    medmfr
) VALUES (
    2,
    'Lisinopril',
    'Capsule',
    'Generic'
);
-- Inserting record 3
INSERT INTO medication (
    medicationid,
    medicationname,
    dosageform,
    medmfr
) VALUES (
    3,
    'Amoxicillin',
    'Liquid',
    'Pfizer'
);
-- Inserting record 4
```

```
INSERT INTO medication (
    medicationid,
    medicationname,
    dosageform,
    medmfr
) VALUES (
    4,
    'Simvastatin',
    'Tablet',
    'Merck'
);
-- Inserting record 5
INSERT INTO medication (
    medicationid,
    medicationname,
    dosageform,
    medmfr
) VALUES (
    5,
    'Omeprazole',
    'Capsule',
    'AstraZeneca'
);
-- Inserting record 6
INSERT INTO medication (
    medicationid,
    medicationname,
    dosageform,
    medmfr
) VALUES (
    6,
    'Metformin',
    'Tablet',
    'Novartis'
);
-- Inserting record 7
INSERT INTO medication (
    medicationid,
    medicationname,
    dosageform,
```

```
    medmfr
) VALUES (
    7,
    'Atorvastatin',
    'Tablet',
    'Pfizer'
);
-- Inserting record 8
INSERT INTO medication (
    medicationid,
    medicationname,
    dosageform,
    medmfr
) VALUES (
    8,
    'Citalopram',
    'Tablet',
    'Generic'
);
-- Inserting record 9
INSERT INTO medication (
    medicationid,
    medicationname,
    dosageform,
    medmfr
) VALUES (
    9,
    'Ibuprofen',
    'Tablet',
    'JohnsonAndJohnson'
);
-- Inserting record 10
INSERT INTO medication (
    medicationid,
    medicationname,
    dosageform,
    medmfr
) VALUES (
    10,
    'Levothyroxine',
```

```
    'Tablet',
    'AbbVie'
);
------administration----
-- Inserting record 1
INSERT INTO administration (
    administrationid,
    adminname,
    admincontact
) VALUES (
    1,
    'AdminName1',
    'Contact1'
);
-- Inserting record 2
INSERT INTO administration (
    administrationid,
    adminname,
    admincontact
) VALUES (
    2,
    'AdminName2',
    'Contact2'
);
-- Inserting record 3
INSERT INTO administration (
    administrationid,
    adminname,
    admincontact
) VALUES (
    3,
    'AdminName3',
    'Contact3'
);
-- Inserting record 4
INSERT INTO administration (
    administrationid,
    adminname,
    admincontact
) VALUES (
```

```
    4,
    'AdminName4',
    'Contact4'
);
-- Inserting record 5
INSERT INTO administration (
    administrationid,
    adminname,
    admincontact
) VALUES (
    5,
    'AdminName5',
    'Contact5'
);
-- Inserting record 6
INSERT INTO administration (
    administrationid,
    adminname,
    admincontact
) VALUES (
    6,
    'AdminName6',
    'Contact6'
);
-- Inserting record 7
INSERT INTO administration (
    administrationid,
    adminname,
    admincontact
) VALUES (
    7,
    'AdminName7',
    'Contact7'
);
-- Inserting record 8
INSERT INTO administration (
    administrationid,
    adminname,
    admincontact
) VALUES (
```

```
    8,
    'AdminName8',
    'Contact8'
);
-- Inserting record 9
INSERT INTO administration (
    administrationid,
    adminname,
    admincontact
) VALUES (
    9,
    'AdminName9',
    'Contact9'
);
-- Inserting record 10
INSERT INTO administration (
    administrationid,
    adminname,
    admincontact
) VALUES (
    10,
    'AdminName10',
    'Contact10'
);
------medicalequipment-----
 -- Inserting record 1
INSERT INTO medicalequipment (
    equipmentid,
    equipmentname,
    manufacturer,
    meddateacquired
) VALUES (
    50,
    'EquipmentName50',
    'Manufacturer50',
    '2023-01-01'
);
-- Inserting record 2
INSERT INTO medicalequipment (
    equipmentid,
```

```
    equipmentname,
    manufacturer,
    meddateacquired
) VALUES (
    2,
    'EquipmentName2',
    'Manufacturer2',
    '2023-02-01'
);
-- Inserting record 3
INSERT INTO medicalequipment (
    equipmentid,
    equipmentname,
    manufacturer,
    meddateacquired
) VALUES (
    3,
    'EquipmentName3',
    'Manufacturer3',
    '2023-03-01'
);
-- Inserting record 4
INSERT INTO medicalequipment (
    equipmentid,
    equipmentname,
    manufacturer,
    meddateacquired
) VALUES (
    4,
    'EquipmentName4',
    'Manufacturer4',
    '2023-04-01'
);
-- Inserting record 5
INSERT INTO medicalequipment (
    equipmentid,
    equipmentname,
    manufacturer,
    meddateacquired
) VALUES (
```

```
    5,
    'EquipmentName5',
    'Manufacturer5',
    '2023-05-01'
);
-- Inserting record 6
INSERT INTO medicalequipment (
    equipmentid,
    equipmentname,
    manufacturer,
    meddateacquired
) VALUES (
    6,
    'EquipmentName6',
    'Manufacturer6',
    '2023-06-01'
);
-- Inserting record 7
INSERT INTO medicalequipment (
    equipmentid,
    equipmentname,
    manufacturer,
    meddateacquired
) VALUES (
    7,
    'EquipmentName7',
    'Manufacturer7',
    '2023-07-01'
);
-- Inserting record 8
INSERT INTO medicalequipment (
    equipmentid,
    equipmentname,
    manufacturer,
    meddateacquired
) VALUES (
    8,
    'EquipmentName8',
    'Manufacturer8',
    '2023-08-01'
```

```
);
-- Inserting record 9
INSERT INTO medicalequipment (
    equipmentid,
    equipmentname,
    manufacturer,
    meddateacquired
) VALUES (
    9,
    'EquipmentName9',
    'Manufacturer9',
    '2023-09-01'
);
-- Inserting record 10
INSERT INTO medicalequipment (
    equipmentid,
    equipmentname,
    manufacturer,
    meddateacquired
) VALUES (
    10,
    'EquipmentName10',
    'Manufacturer10',
    '2023-10-01'
);
---healthcareprovider----
-- Inserting record 1
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    1,
    1,
    'HPLName1',
    'Speciality1',
    'Contact1'
);
```

```
-- Inserting record 2
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    2,
    8,
    'HPLName2',
    'Speciality2',
    'Contact2'
);
-- Inserting record 3
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    3,
    1,
    'HPLName3',
    'Speciality3',
    'Contact3'
);
-- Inserting record 4
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    4,
    1,
    'HPLName4',
    'Speciality4',
```

```
    'Contact4'
);
-- Inserting record 5
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    5,
    1,
    'HPLName5',
    'Speciality5',
    'Contact5'
);
-- Inserting record 6
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    6,
    2,
    'HPLName6',
    'Speciality6',
    'Contact6'
);
-- Inserting record 7
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    7,
    2,
```

```
    'HPLName7',
    'Speciality7',
    'Contact7'
);
-- Inserting record 8
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    8,
    9,
    'HPLName8',
    'Speciality8',
    'Contact8'
);
-- Inserting record 9
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    9,
    2,
    'HPLName9',
    'Speciality9',
    'Contact9'
);
-- Inserting record 10
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
```

```
    10,
    2,
    'HPLName10',
    'Speciality10',
    'Contact10'
);
-- Inserting record 11
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    11,
    10,
    'HPLName11',
    'Speciality11',
    'Contact11'
);
-- Inserting record 12
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    12,
    3,
    'HPLName12',
    'Speciality12',
    'Contact12'
);
-- Inserting record 13
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
```

```
    healthcarecontact
) VALUES (
    13,
    3,
    'HPLName13',
    'Speciality13',
    'Contact13'
);
-- Inserting record 14
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    14,
    3,
    'HPLName14',
    'Speciality14',
    'Contact14'
);
-- Inserting record 15
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    15,
    6,
    'HPLName15',
    'Speciality15',
    'Contact15'
);
-- Inserting record 16
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
```

```
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    16,
    4,
    'HPLName16',
    'Speciality16',
    'Contact16'
);
-- Inserting record 17
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    17,
    4,
    'HPLName17',
    'Speciality17',
    'Contact17'
);
-- Inserting record 18
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    18,
    4,
    'HPLName18',
    'Speciality18',
    'Contact18'
);
-- Inserting record 19
INSERT INTO healthcareprovider (
```

```
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    19,
    4,
    'HPLName19',
    'Speciality19',
    'Contact19'
);
-- Inserting record 20
INSERT INTO healthcareprovider (
    providerid,
    administrationid,
    hplname,
    speciality,
    healthcarecontact
) VALUES (
    20,
    5,
    'HPLName20',
    'Speciality20',
    'Contact20'
);
----doctor------
-- Inserting record 1
INSERT INTO doctor (
    doctorid,
    providerid,
    docfname,
    doclname,
    docemail,
    doccontact
) VALUES (
    1,
    1,
    'John',
    'Doe',
```

```
    'john.doe@example.com',
    'Contact1'
);
-- Inserting record 2
INSERT INTO doctor (
    doctorid,
    providerid,
    docfname,
    doclname,
    docemail,
    doccontact
) VALUES (
    2,
    1,
    'Jane',
    'Smith',
    'jane.smith@example.com',
    'Contact2'
);
-- Inserting record 3
INSERT INTO doctor (
    doctorid,
    providerid,
    docfname,
    doclname,
    docemail,
    doccontact
) VALUES (
    3,
    1,
    'Michael',
    'Johnson',
    'michael.johnson@example.com',
    'Contact3'
);
-- Inserting record 4
INSERT INTO doctor (
    doctorid,
    providerid,
    docfname,
```

```
    doclname,
    docemail,
    doccontact
) VALUES (
    4,
    2,
    'Emily',
    'Williams',
    'emily.williams@example.com',
    'Contact4'
);
-- Inserting record 5
INSERT INTO doctor (
    doctorid,
    providerid,
    docfname,
    doclname,
    docemail,
    doccontact
) VALUES (
    5,
    2,
    'Daniel',
    'Jones',
    'daniel.jones@example.com',
    'Contact5'
);
-- Inserting record 6
INSERT INTO doctor (
    doctorid,
    providerid,
    docfname,
    doclname,
    docemail,
    doccontact
) VALUES (
    6,
    2,
    'Olivia',
    'Brown',
```

```
    'olivia.brown@example.com',
    'Contact6'
);
-- Inserting record 7
INSERT INTO doctor (
    doctorid,
    providerid,
    docfname,
    doclname,
    docemail,
    doccontact
) VALUES (
    7,
    3,
    'Liam',
    'Miller',
    'liam.miller@example.com',
    'Contact7'
);
-- Inserting record 8
INSERT INTO doctor (
    doctorid,
    providerid,
    docfname,
    doclname,
    docemail,
    doccontact
) VALUES (
    8,
    3,
    'Ava',
    'Davis',
    'ava.davis@example.com',
    'Contact8'
);
-- Inserting record 9
INSERT INTO doctor (
    doctorid,
    providerid,
    docfname,
```

```
    doclname,
    docemail,
    doccontact
) VALUES (
    9,
    3,
    'Lucas',
    'Garcia',
    'lucas.garcia@example.com',
    'Contact9'
);
-- Inserting record 10
INSERT INTO doctor (
    doctorid,
    providerid,
    docfname,
    doclname,
    docemail,
    doccontact
) VALUES (
    10,
    4,
    'Sophia',
    'Rodriguez',
    'sophia.rodriguez@example.com',
    'Contact10'
);
---insurance provider----
-- Inserting record 1
INSERT INTO insuranceprovider (
    insuranceid,
    administrationid,
    insurancename,
    insurancecontact
) VALUES (
    1,
    1,
    'InsuranceProvider1',
    'Contact1'
);
```

```
-- Inserting record 2
INSERT INTO insuranceprovider (
    insuranceid,
    administrationid,
    insurancename,
    insurancecontact
) VALUES (
    2,
    1,
    'InsuranceProvider2',
    'Contact2'
);
-- Inserting record 3
INSERT INTO insuranceprovider (
    insuranceid,
    administrationid,
    insurancename,
    insurancecontact
) VALUES (
    3,
    1,
    'InsuranceProvider3',
    'Contact3'
);
-- Inserting record 4
INSERT INTO insuranceprovider (
    insuranceid,
    administrationid,
    insurancename,
    insurancecontact
) VALUES (
    4,
    2,
    'InsuranceProvider4',
    'Contact4'
);
-- Inserting record 5
INSERT INTO insuranceprovider (
    insuranceid,
    administrationid,
```

```
    insurancename,
    insurancecontact
) VALUES (
    5,
    2,
    'InsuranceProvider5',
    'Contact5'
);
-- Inserting record 6
INSERT INTO insuranceprovider (
    insuranceid,
    administrationid,
    insurancename,
    insurancecontact
) VALUES (
    6,
    2,
    'InsuranceProvider6',
    'Contact6'
);
-- Inserting record 7
INSERT INTO insuranceprovider (
    insuranceid,
    administrationid,
    insurancename,
    insurancecontact
) VALUES (
    7,
    3,
    'InsuranceProvider7',
    'Contact7'
);
-- Inserting record 8
INSERT INTO insuranceprovider (
    insuranceid,
    administrationid,
    insurancename,
    insurancecontact
) VALUES (
    8,
```

```
    3,
    'InsuranceProvider8',
    'Contact8'
);
-- Inserting record 9
INSERT INTO insuranceprovider (
    insuranceid,
    administrationid,
    insurancename,
    insurancecontact
) VALUES (
    9,
    3,
    'InsuranceProvider9',
    'Contact9'
);
-- Inserting record 10
INSERT INTO insuranceprovider (
    insuranceid,
    administrationid,
    insurancename,
    insurancecontact
) VALUES (
    10,
    4,
    'InsuranceProvider10',
    'Contact10'
);
------patient-----
-- Inserting record 1
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
```

```
) VALUES (
    1,
    1,
    1,
    'Alice',
    'Johnson',
    '1990-05-15',
    'F',
    'Contact1',
    'Address1'
);
-- Inserting record 2
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    2,
    1,
    2,
    'Bob',
    'Smith',
    '1985-08-20',
    'M',
    'Contact2',
    'Address2'
);
-- Inserting record 3
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
```

```
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    3,
    1,
    3,
    'Catherine',
    'Davis',
    '1992-03-10',
    'F',
    'Contact3',
    'Address3'
);
-- Inserting record 4
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    4,
    2,
    1,
    'David',
    'Wilson',
    '1988-12-05',
    'M',
    'Contact4',
    'Address4'
);
-- Inserting record 5
INSERT INTO patient (
    patientid,
```

```
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    5,
    2,
    2,
    'Emma',
    'Jones',
    '1995-01-25',
    'F',
    'Contact5',
    'Address5'
);
-- Inserting record 6
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    6,
    2,
    3,
    'Frank',
    'Brown',
    '1980-07-18',
    'M',
    'Contact6',
    'Address6'
```

```
);
-- Inserting record 7
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    7,
    3,
    1,
    'Grace',
    'Miller',
    '1998-09-30',
    'F',
    'Contact7',
    'Address7'
);
-- Inserting record 8
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    8,
    3,
    2,
    'Henry',
    'Garcia',
```

```
    '1987-06-12',
    'M',
    'Contact8',
    'Address8'
);
-- Inserting record 9
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    9,
    3,
    3,
    'Isabella',
    'Taylor',
    '1994-11-08',
    'F',
    'Contact9',
    'Address9'
);
-- Inserting record 10
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    10,
```

```
    4,
    1,
    'Jack',
    'Rodriguez',
    '1983-04-03',
    'M',
    'Contact10',
    'Address10'
);
-- Inserting record 11
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    11,
    4,
    2,
    'Katherine',
    'Lopez',
    '1991-02-14',
    'F',
    'Contact11',
    'Address11'
);
-- Inserting record 12
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
```

```
    pcontact,
    paddress
) VALUES (
    12,
    4,
    3,
    'Liam',
    'Martinez',
    '1989-07-22',
    'M',
    'Contact12',
    'Address12'
);
-- Inserting record 13
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    13,
    5,
    1,
    'Mia',
    'Hernandez',
    '1996-10-05',
    'F',
    'Contact13',
    'Address13'
);
-- Inserting record 14
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
```

```
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    14,
    5,
    2,
    'Noah',
    'Flores',
    '1986-08-29',
    'M',
    'Contact14',
    'Address14'
);
-- Inserting record 15
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    15,
    5,
    3,
    'Olivia',
    'Perez',
    '1993-12-17',
    'F',
    'Contact15',
    'Address15'
);
-- Inserting record 16
```

```
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    16,
    6,
    1,
    'Patrick',
    'Sanchez',
    '1984-06-08',
    'M',
    'Contact16',
    'Address16'
);
-- Inserting record 17
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    17,
    6,
    2,
    'Quinn',
    'Rivera',
    '1997-03-01',
    'F',
```

```
    'Contact17',
    'Address17'
);
-- Inserting record 18
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    18,
    6,
    3,
    'Ryan',
    'Torres',
    '1982-09-14',
    'M',
    'Contact18',
    'Address18'
);
-- Inserting record 19
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    19,
    7,
    1,
```

```
    'Sophia',
    'Gomez',
    '1999-04-27',
    'F',
    'Contact19',
    'Address19'
);
-- Inserting record 20
INSERT INTO patient (
    patientid,
    doctorid,
    insuranceid,
    patientfname,
    patientlname,
    patientdob,
    pgender,
    pcontact,
    paddress
) VALUES (
    20,
    7,
    2,
    'Tyler',
    'Diaz',
    '1981-11-10',
    'M',
    'Contact20',
    'Address20'
);
------integrated department---
-- Inserting record 1
INSERT INTO integratedcaredepartment (
    icdepartmentid,
    providerid,
    departmentcontact
) VALUES (
    1,
    1,
    'Contact1'
);
```

```
-- Inserting record 2
INSERT INTO integratedcaredepartment (
    icdepartmentid,
    providerid,
    departmentcontact
) VALUES (
    2,
    1,
    'Contact2'
);
-- Inserting record 3
INSERT INTO integratedcaredepartment (
    icdepartmentid,
    providerid,
    departmentcontact
) VALUES (
    3,
    1,
    'Contact3'
);
-- Inserting record 4
INSERT INTO integratedcaredepartment (
    icdepartmentid,
    providerid,
    departmentcontact
) VALUES (
    4,
    2,
    'Contact4'
);
-- Inserting record 5
INSERT INTO integratedcaredepartment (
    icdepartmentid,
    providerid,
    departmentcontact
) VALUES (
    5,
    2,
    'Contact5'
);
```

```
-- Inserting record 6
INSERT INTO integratedcaredepartment (
    icdepartmentid,
    providerid,
    departmentcontact
) VALUES (
    6,
    2,
    'Contact6'
);
-- Inserting record 7
INSERT INTO integratedcaredepartment (
    icdepartmentid,
    providerid,
    departmentcontact
) VALUES (
    7,
    3,
    'Contact7'
);
-- Inserting record 8
INSERT INTO integratedcaredepartment (
    icdepartmentid,
    providerid,
    departmentcontact
) VALUES (
    8,
    3,
    'Contact8'
);
-- Inserting record 9
INSERT INTO integratedcaredepartment (
    icdepartmentid,
    providerid,
    departmentcontact
) VALUES (
    9,
    3,
    'Contact9'
);
```

```
-- Inserting record 10
INSERT INTO integratedcaredepartment (
    icdepartmentid,
    providerid,
    departmentcontact
) VALUES (
    10,
    4,
    'Contact10'
);
------department----
-- Inserting record 1
INSERT INTO department (
    departmentid,
    icdepartmentid,
    departmentname,
    departmentcontact
) VALUES (
    1,
    1,
    'Department1',
    'Contact1'
);
-- Inserting record 2
INSERT INTO department (
    departmentid,
    icdepartmentid,
    departmentname,
    departmentcontact
) VALUES (
    2,
    1,
    'Department2',
    'Contact2'
);
-- Inserting record 3
INSERT INTO department (
    departmentid,
    icdepartmentid,
    departmentname,
```

```
    departmentcontact
) VALUES (
    3,
    1,
    'Department3',
    'Contact3'
);
-- Inserting record 4
INSERT INTO department (
    departmentid,
    icdepartmentid,
    departmentname,
    departmentcontact
) VALUES (
    4,
    2,
    'Department4',
    'Contact4'
);
-- Inserting record 5
INSERT INTO department (
    departmentid,
    icdepartmentid,
    departmentname,
    departmentcontact
) VALUES (
    5,
    2,
    'Department5',
    'Contact5'
);
-- Inserting record 6
INSERT INTO department (
    departmentid,
    icdepartmentid,
    departmentname,
    departmentcontact
) VALUES (
    6,
    2,
```

```
    'Department6',
    'Contact6'
);
-- Inserting record 7
INSERT INTO department (
    departmentid,
    icdepartmentid,
    departmentname,
    departmentcontact
) VALUES (
    7,
    3,
    'Department7',
    'Contact7'
);
-- Inserting record 8
INSERT INTO department (
    departmentid,
    icdepartmentid,
    departmentname,
    departmentcontact
) VALUES (
    8,
    3,
    'Department8',
    'Contact8'
);
-- Inserting record 9
INSERT INTO department (
    departmentid,
    icdepartmentid,
    departmentname,
    departmentcontact
) VALUES (
    9,
    3,
    'Department9',
    'Contact9'
);
-- Inserting record 10
```

```
INSERT INTO department (
    departmentid,
    icdepartmentid,
    departmentname,
    departmentcontact
) VALUES (
    10,
    4,
    'Department10',
    'Contact10'
);
---------laboratory------
-- Inserting record 1
INSERT INTO laboratory (
    labid,
    departmentid,
    equipmentid,
    laboratoriescontact,
    labname
) VALUES (
    1,
    1,
    1,
    'LabContact1',
    'LabName1'
);
-- Inserting record 2
INSERT INTO laboratory (
    labid,
    departmentid,
    equipmentid,
    laboratoriescontact,
    labname
) VALUES (
    2,
    1,
    2,
    'LabContact2',
    'LabName2'
);
```

```
-- Inserting record 3
INSERT INTO laboratory (
    labid,
    departmentid,
    equipmentid,
    laboratoriescontact,
    labname
) VALUES (
    3,
    1,
    3,
    'LabContact3',
    'LabName3'
);
-- Inserting record 4
INSERT INTO laboratory (
    labid,
    departmentid,
    equipmentid,
    laboratoriescontact,
    labname
) VALUES (
    4,
    2,
    1,
    'LabContact4',
    'LabName4'
);
-- Inserting record 5
INSERT INTO laboratory (
    labid,
    departmentid,
    equipmentid,
    laboratoriescontact,
    labname
) VALUES (
    5,
    2,
    2,
    'LabContact5',
```

```
    'LabName5'
);
-- Inserting record 6
INSERT INTO laboratory (
    labid,
    departmentid,
    equipmentid,
    laboratoriescontact,
    labname
) VALUES (
    6,
    2,
    3,
    'LabContact6',
    'LabName6'
);
-- Inserting record 7
INSERT INTO laboratory (
    labid,
    departmentid,
    equipmentid,
    laboratoriescontact,
    labname
) VALUES (
    7,
    3,
    1,
    'LabContact7',
    'LabName7'
);
-- Inserting record 8
INSERT INTO laboratory (
    labid,
    departmentid,
    equipmentid,
    laboratoriescontact,
    labname
) VALUES (
    8,
    3,
```

```
    2,
    'LabContact8',
    'LabName8'
);
-- Inserting record 9
INSERT INTO laboratory (
    labid,
    departmentid,
    equipmentid,
    laboratoriescontact,
    labname
) VALUES (
    9,
    3,
    3,
    'LabContact9',
    'LabName9'
);
-- Inserting record 10
INSERT INTO laboratory (
    labid,
    departmentid,
    equipmentid,
    laboratoriescontact,
    labname
) VALUES (
    10,
    4,
    1,
    'LabContact10',
    'LabName10'
);
-------appointment----
-- Inserting record 1
INSERT INTO appointment (
    appointmentid,
    patientid,
    providerid,
    appdateandtime,
    apptpurpose
```

```
) VALUES (
    1,
    1,
    1,
    TO_DATE('2023-01-15 10:00', 'YYYY-MM-DD HH24:MI'),
    'Checkup'
);
-- Inserting record 2
INSERT INTO appointment (
    appointmentid,
    patientid,
    providerid,
    appdateandtime,
    apptpurpose
) VALUES (
    2,
    2,
    1,
    TO_DATE('2023-02-20 14:30', 'YYYY-MM-DD HH24:MI'),
    'Follow-up'
);
-- Inserting record 3
INSERT INTO appointment (
    appointmentid,
    patientid,
    providerid,
    appdateandtime,
    apptpurpose
) VALUES (
    3,
    3,
    2,
    TO_DATE('2023-03-10 11:45', 'YYYY-MM-DD HH24:MI'),
    'Consultation'
);
-- Inserting record 4
INSERT INTO appointment (
    appointmentid,
    patientid,
    providerid,
```

```
    appdateandtime,
    apptpurpose
) VALUES (
    4,
    4,
    2,
    TO_DATE('2023-04-05 09:15', 'YYYY-MM-DD HH24:MI'),
    'Checkup'
);
-- Inserting record 5
INSERT INTO appointment (
    appointmentid,
    patientid,
    providerid,
    appdateandtime,
    apptpurpose
) VALUES (
    5,
    5,
    3,
    TO_DATE('2023-05-25 13:00', 'YYYY-MM-DD HH24:MI'),
    'Follow-up'
);
-- Inserting record 6
INSERT INTO appointment (
    appointmentid,
    patientid,
    providerid,
    appdateandtime,
    apptpurpose
) VALUES (
    6,
    6,
    3,
    TO_DATE('2023-06-18 15:30', 'YYYY-MM-DD HH24:MI'),
    'Consultation'
);
-- Inserting record 7
INSERT INTO appointment (
    appointmentid,
```

```
    patientid,
    providerid,
    appdateandtime,
    apptpurpose
) VALUES (
    7,
    7,
    4,
    TO_DATE('2023-07-30 10:45', 'YYYY-MM-DD HH24:MI'),
    'Checkup'
);
-- Inserting record 8
INSERT INTO appointment (
    appointmentid,
    patientid,
    providerid,
    appdateandtime,
    apptpurpose
) VALUES (
    8,
    8,
    4,
    TO_DATE('2023-08-15 14:00', 'YYYY-MM-DD HH24:MI'),
    'Follow-up'
);
-- Inserting record 9
INSERT INTO appointment (
    appointmentid,
    patientid,
    providerid,
    appdateandtime,
    apptpurpose
) VALUES (
    9,
    9,
    5,
    TO_DATE('2023-09-05 12:30', 'YYYY-MM-DD HH24:MI'),
    'Consultation'
);
-- Inserting record 10
```

```
INSERT INTO appointment (
    appointmentid,
    patientid,
    providerid,
    appdateandtime,
    apptpurpose
) VALUES (
    10,
    10,
    5,
    TO_DATE('2023-10-20 09:00', 'YYYY-MM-DD HH24:MI'),
    'Checkup'
);
-----prescription----
-- Inserting record 1
INSERT INTO prescription (
    prescriptionid,
    patientid,
    doctorid,
    dosage,
    dateprescribed
) VALUES (
    1,
    1,
    1,
    '10mg once a day',
    TO_DATE('2023-01-15', 'YYYY-MM-DD')
);
-- Inserting record 2
INSERT INTO prescription (
    prescriptionid,
    patientid,
    doctorid,
    dosage,
    dateprescribed
) VALUES (
    2,
    1,
    2,
    '20mg twice a day',
```

```
    TO_DATE('2023-02-20', 'YYYY-MM-DD')
);
-- Inserting record 3
INSERT INTO prescription (
    prescriptionid,
    patientid,
    doctorid,
    dosage,
    dateprescribed
) VALUES (
    3,
    3,
    3,
    '5mg three times a day',
    TO_DATE('2023-03-10', 'YYYY-MM-DD')
);
-- Inserting record 4
INSERT INTO prescription (
    prescriptionid,
    patientid,
    doctorid,
    dosage,
    dateprescribed
) VALUES (
    4,
    4,
    3,
    '15mg once a day',
    TO_DATE('2023-04-05', 'YYYY-MM-DD')
);
-- Inserting record 5
INSERT INTO prescription (
    prescriptionid,
    patientid,
    doctorid,
    dosage,
    dateprescribed
) VALUES (
    5,
    5,
```

```
    5,
    '25mg twice a day',
    TO_DATE('2023-05-25', 'YYYY-MM-DD')
);
-- Inserting record 6
INSERT INTO prescription (
    prescriptionid,
    patientid,
    doctorid,
    dosage,
    dateprescribed
) VALUES (
    6,
    6,
    6,
    '10mg three times a day',
    TO_DATE('2023-06-18', 'YYYY-MM-DD')
);
-- Inserting record 7
INSERT INTO prescription (
    prescriptionid,
    patientid,
    doctorid,
    dosage,
    dateprescribed
) VALUES (
    7,
    7,
    7,
    '30mg once a day',
    TO_DATE('2023-07-30', 'YYYY-MM-DD')
);
-- Inserting record 8
INSERT INTO prescription (
    prescriptionid,
    patientid,
    doctorid,
    dosage,
    dateprescribed
) VALUES (
```

```
    8,
    8,
    8,
    '15mg twice a day',
    TO_DATE('2023-08-15', 'YYYY-MM-DD')
);
-- Inserting record 9
INSERT INTO prescription (
    prescriptionid,
    patientid,
    doctorid,
    dosage,
    dateprescribed
) VALUES (
    9,
    9,
    9,
    '20mg three times a day',
    TO_DATE('2023-09-05', 'YYYY-MM-DD')
);
-- Inserting record 10
INSERT INTO prescription (
    prescriptionid,
    patientid,
    doctorid,
    dosage,
    dateprescribed
) VALUES (
    10,
    10,
    10,
    '10mg once a day',
    TO_DATE('2023-10-20', 'YYYY-MM-DD')
);
---dischare summary---
-- Inserting record 1
INSERT INTO dischargesummary (
    dischargeid,
    patientid,
    dateofdischarge,
```

```
    summary
) VALUES (
    1,
    5,
    TO_DATE('2023-01-15', 'YYYY-MM-DD'),
    'Patient was discharged after successful treatment.'
);
-- Inserting record 2
INSERT INTO dischargesummary (
    dischargeid,
    patientid,
    dateofdischarge,
    summary
) VALUES (
    2,
    2,
    TO_DATE('2023-02-20', 'YYYY-MM-DD'),
    'Patient recovered well and was discharged.'
);
-- Inserting record 3
INSERT INTO dischargesummary (
    dischargeid,
    patientid,
    dateofdischarge,
    summary
) VALUES (
    3,
    3,
    TO_DATE('2023-03-10', 'YYYY-MM-DD'),
    'Discharge summary for patient.'
);
-- Inserting record 4
INSERT INTO dischargesummary (
    dischargeid,
    patientid,
    dateofdischarge,
    summary
) VALUES (
    4,
    9,
```

```
    TO_DATE('2023-04-05', 'YYYY-MM-DD'),
    'Patient discharged with post-operative care instructions.'
);
-- Inserting record 8
INSERT INTO dischargesummary (
    dischargeid,
    patientid,
    dateofdischarge,
    summary
) VALUES (
    8,
    8,
    TO_DATE('2023-08-15', 'YYYY-MM-DD'),
    'Patient released from the hospital.'
);
-- Inserting record 9
INSERT INTO dischargesummary (
    dischargeid,
    patientid,
    dateofdischarge,
    summary
) VALUES (
    9,
    9,
    TO_DATE('2023-09-05', 'YYYY-MM-DD'),
    'Detailed discharge summary for the patient.'
);
-- Inserting record 10
INSERT INTO dischargesummary (
    dischargeid,
    patientid,
    dateofdischarge,
    summary
) VALUES (
    10,
    3,
    TO_DATE('2023-10-20', 'YYYY-MM-DD'),
    'Patient successfully discharged after treatment.'
);
-----medicalrecord----
```

```
-- Inserting record 1
INSERT INTO medicalrecord (
    recordid,
    patientid,
    diagnosis,
    dateofvisit,
    notes
) VALUES (
    1,
    1,
    'Common cold',
    TO_DATE('2023-01-15', 'YYYY-MM-DD'),
    'Prescribed rest and fluids.'
);
-- Inserting record 2
INSERT INTO medicalrecord (
    recordid,
    patientid,
    diagnosis,
    dateofvisit,
    notes
) VALUES (
    2,
    2,
    'Fractured arm',
    TO_DATE('2023-02-20', 'YYYY-MM-DD'),
    'Applied cast and prescribed pain medication.'
);
-- Inserting record 3
INSERT INTO medicalrecord (
    recordid,
    patientid,
    diagnosis,
    dateofvisit,
    notes
) VALUES (
    3,
    3,
    'Allergies',
    TO_DATE('2023-03-10', 'YYYY-MM-DD'),
```

```
    'Prescribed antihistamines.'
);
-- Inserting record 4
INSERT INTO medicalrecord (
    recordid,
    patientid,
    diagnosis,
    dateofvisit,
    notes
) VALUES (
    4,
    4,
    'Appendicitis',
    TO_DATE('2023-04-05', 'YYYY-MM-DD'),
    'Performed appendectomy.'
);
-- Inserting record 5
INSERT INTO medicalrecord (
    recordid,
    patientid,
    diagnosis,
    dateofvisit,
    notes
) VALUES (
    5,
    5,
    'High blood pressure',
    TO_DATE('2023-05-25', 'YYYY-MM-DD'),
    'Prescribed medication and advised lifestyle changes.'
);
-- Inserting record 6
INSERT INTO medicalrecord (
    recordid,
    patientid,
    diagnosis,
    dateofvisit,
    notes
) VALUES (
    6,
    6,
```

```
    'Diabetes',
    TO_DATE('2023-06-18', 'YYYY-MM-DD'),
    'Started insulin therapy and dietary recommendations.'
);
-- Inserting record 7
INSERT INTO medicalrecord (
    recordid,
    patientid,
    diagnosis,
    dateofvisit,
    notes
) VALUES (
    7,
    7,
    'Influenza',
    TO_DATE('2023-07-30', 'YYYY-MM-DD'),
    'Prescribed antiviral medication and bed rest.'
);
-- Inserting record 8
INSERT INTO medicalrecord (
    recordid,
    patientid,
    diagnosis,
    dateofvisit,
    notes
) VALUES (
    8,
    8,
    'Sprained ankle',
    TO_DATE('2023-08-15', 'YYYY-MM-DD'),
    'Applied ice pack and recommended rest.'
);
-- Inserting record 9
INSERT INTO medicalrecord (
    recordid,
    patientid,
    diagnosis,
    dateofvisit,
    notes
) VALUES (
```

```
    9,
    9,
    'Migraine',
    TO_DATE('2023-09-05', 'YYYY-MM-DD'),
    'Prescribed pain relievers and advised stress management.'
);
-- Inserting record 10
INSERT INTO medicalrecord (
    recordid,
    patientid,
    diagnosis,
    dateofvisit,
    notes
) VALUES (
    10,
    10,
    'Gastritis',
    TO_DATE('2023-10-20', 'YYYY-MM-DD'),
    'Prescribed antacids and dietary modifications.'
);
------surgery----
-- Inserting record 1
INSERT INTO surgery (
    surgeryid,
    patientid,
    doctorid,
    surgerytype,
    surgerydate
) VALUES (
    1,
    1,
    1,
    'Appendectomy',
    TO_DATE('2023-01-15', 'YYYY-MM-DD')
);
-- Inserting record 2
INSERT INTO surgery (
    surgeryid,
    patientid,
    doctorid,
```

```
    surgerytype,
    surgerydate
) VALUES (
    2,
    2,
    2,
    'Fracture Repair',
    TO_DATE('2023-02-20', 'YYYY-MM-DD')
);
-- Inserting record 3
INSERT INTO surgery (
    surgeryid,
    patientid,
    doctorid,
    surgerytype,
    surgerydate
) VALUES (
    3,
    3,
    3,
    'Allergy Testing',
    TO_DATE('2023-03-10', 'YYYY-MM-DD')
);
-- Inserting record 4
INSERT INTO surgery (
    surgeryid,
    patientid,
    doctorid,
    surgerytype,
    surgerydate
) VALUES (
    4,
    4,
    4,
    'Appendectomy',
    TO_DATE('2023-04-05', 'YYYY-MM-DD')
);
-- Inserting record 5
INSERT INTO surgery (
    surgeryid,
```

```
    patientid,
    doctorid,
    surgerytype,
    surgerydate
) VALUES (
    5,
    5,
    5,
    'Cardiac Bypass',
    TO_DATE('2023-05-25', 'YYYY-MM-DD')
);
-- Inserting record 6
INSERT INTO surgery (
    surgeryid,
    patientid,
    doctorid,
    surgerytype,
    surgerydate
) VALUES (
    6,
    6,
    6,
    'Knee Replacement',
    TO_DATE('2023-06-18', 'YYYY-MM-DD')
);
-- Inserting record 7
INSERT INTO surgery (
    surgeryid,
    patientid,
    doctorid,
    surgerytype,
    surgerydate
) VALUES (
    7,
    7,
    7,
    'Appendectomy',
    TO_DATE('2023-07-30', 'YYYY-MM-DD')
);
-- Inserting record 8
```

```
INSERT INTO surgery (
    surgeryid,
    patientid,
    doctorid,
    surgerytype,
    surgerydate
) VALUES (
    8,
    8,
    8,
    'Appendectomy',
    TO_DATE('2023-08-15', 'YYYY-MM-DD')
);
-- Inserting record 9
INSERT INTO surgery (
    surgeryid,
    patientid,
    doctorid,
    surgerytype,
    surgerydate
) VALUES (
    9,
    9,
    9,
    'Spinal Fusion',
    TO_DATE('2023-09-05', 'YYYY-MM-DD')
);
-- Inserting record 10
INSERT INTO surgery (
    surgeryid,
    patientid,
    doctorid,
    surgerytype,
    surgerydate
) VALUES (
    10,
    10,
    10,
    'Appendectomy',
    TO_DATE('2023-10-20', 'YYYY-MM-DD')
```

```
);
-----billing----
-- Inserting record 1
INSERT INTO billing (
    billid,
    patientid,
    providerid,
    billamount,
    billdateissued
) VALUES (
    1,
    1,
    1,
    500.00,
    TO_DATE('2023-01-15', 'YYYY-MM-DD')
);
-- Inserting record 2
INSERT INTO billing (
    billid,
    patientid,
    providerid,
    billamount,
    billdateissued
) VALUES (
    2,
    2,
    2,
    750.00,
    TO_DATE('2023-02-20', 'YYYY-MM-DD')
);
-- Inserting record 3
INSERT INTO billing (
    billid,
    patientid,
    providerid,
    billamount,
    billdateissued
) VALUES (
    3,
    3,
```

```
    3,
    300.00,
    TO_DATE('2023-03-10', 'YYYY-MM-DD')
);
-- Inserting record 4
INSERT INTO billing (
    billid,
    patientid,
    providerid,
    billamount,
    billdateissued
) VALUES (
    4,
    4,
    4,
    1200.00,
    TO_DATE('2023-04-05', 'YYYY-MM-DD')
);
-- Inserting record 5
INSERT INTO billing (
    billid,
    patientid,
    providerid,
    billamount,
    billdateissued
) VALUES (
    5,
    5,
    5,
    1000.00,
    TO_DATE('2023-05-25', 'YYYY-MM-DD')
);
-- Inserting record 6
INSERT INTO billing (
    billid,
    patientid,
    providerid,
    billamount,
    billdateissued
) VALUES (
```

```
    6,
    6,
    6,
    450.00,
    TO_DATE('2023-06-18', 'YYYY-MM-DD')
);
-- Inserting record 7
INSERT INTO billing (
    billid,
    patientid,
    providerid,
    billamount,
    billdateissued
) VALUES (
    7,
    7,
    7,
    600.00,
    TO_DATE('2023-07-30', 'YYYY-MM-DD')
);
-- Inserting record 8
INSERT INTO billing (
    billid,
    patientid,
    providerid,
    billamount,
    billdateissued
) VALUES (
    8,
    8,
    8,
    800.00,
    TO_DATE('2023-08-15', 'YYYY-MM-DD')
);
-- Inserting record 9
INSERT INTO billing (
    billid,
    patientid,
    providerid,
    billamount,
```

```
    billdateissued
) VALUES (
    9,
    9,
    9,
    900.00,
    TO_DATE('2023-09-05', 'YYYY-MM-DD')
);
-- Inserting record 10
INSERT INTO billing (
    billid,
    patientid,
    providerid,
    billamount,
    billdateissued
) VALUES (
    10,
    10,
    10,
    350.00,
    TO_DATE('2023-10-20', 'YYYY-MM-DD')
);
---allergy-----
-- Inserting record 1
INSERT INTO allergy (
    allergyid,
    patientid,
    allergen,
    reaction
) VALUES (
    1,
    1,
    'Pollen',
    'Sneezing and itchy eyes.'
);
-- Inserting record 2
INSERT INTO allergy (
    allergyid,
    patientid,
    allergen,
```

```
    reaction
) VALUES (
    2,
    2,
    'Peanuts',
    'Swelling and difficulty breathing.'
);
-- Inserting record 3
INSERT INTO allergy (
    allergyid,
    patientid,
    allergen,
    reaction
) VALUES (
    3,
    3,
    'Penicillin',
    'Rash and itching.'
);
-- Inserting record 4
INSERT INTO allergy (
    allergyid,
    patientid,
    allergen,
    reaction
) VALUES (
    4,
    4,
    'Dust Mites',
    'Coughing and shortness of breath.'
);
-- Inserting record 5
INSERT INTO allergy (
    allergyid,
    patientid,
    allergen,
    reaction
) VALUES (
    5,
    3,
```

```
    'Shellfish',
    'Nausea and hives.'
);
-- Inserting record 6
INSERT INTO allergy (
    allergyid,
    patientid,
    allergen,
    reaction
) VALUES (
    6,
    6,
    'Mold',
    'Headache and fatigue.'
);
-- Inserting record 7
INSERT INTO allergy (
    allergyid,
    patientid,
    allergen,
    reaction
) VALUES (
    7,
    7,
    'Eggs',
    'Abdominal pain and vomiting.'
);
-- Inserting record 8
INSERT INTO allergy (
    allergyid,
    patientid,
    allergen,
    reaction
) VALUES (
    8,
    9,
    'Latex',
    'Skin rash and itching.'
);
-- Inserting record 9
```

```
INSERT INTO allergy (
    allergyid,
    patientid,
    allergen,
    reaction
) VALUES (
    9,
    9,
    'Insect Bites',
    'Swelling and redness.'
);
-- Inserting record 10
INSERT INTO allergy (
    allergyid,
    patientid,
    allergen,
    reaction
) VALUES (
    10,
    5,
    'Milk',
    'Diarrhea and stomach cramps.'
);
------END-----
```

13. DCL Scripts

In the Public Health Ontario Systems Database project, Data Control Language (DCL) queries play a crucial role in governing and managing access to the database. DCL queries are instrumental in defining and regulating the permissions and privileges granted to users and roles within the system.
Through the use of DCL GRANT queries, specific user roles, such as healthcare providers, administrators, and other authorized personnel, are endowed with the necessary permissions to execute specific actions, ensuring that they can interact with the database in a manner aligned with their responsibilities.
Conversely, DCL REVOKE queries are employed to retract or limit certain permissions, enhancing data security and enforcing access control policies. The careful orchestration of DCL queries in the project ensures that only authorized individuals can perform essential operations such as updating patient records, scheduling appointments, or accessing sensitive medical information.

By leveraging DCL queries effectively, the Public Health Ontario Systems Database project maintains a robust security framework, safeguarding patient confidentiality and upholding regulatory compliance standards within the healthcare domain.

We have attached the script below which includes all the DCL queries performed as part of the project.

DCL Scripts Queries

```
---START----
-----access control----
-- Creating roles
CREATE ROLE healthcare_provider_role;
CREATE ROLE doctor_role;
CREATE ROLE patient_role;
CREATE ROLE insurance_provider_role;
CREATE ROLE admin_role;
-- Granting access to roles for Medication table
GRANT SELECT ON medication TO healthcare_provider_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON medication TO admin_role;
-- Granting access to roles for Administration table
GRANT SELECT, INSERT, UPDATE, DELETE ON administration TO admin_role;
-- Granting access to roles for Medicalequipment table
GRANT SELECT ON medicalequipment TO healthcare_provider_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON medicalequipment TO admin_role;
-- Granting access to roles for Healthcareprovider table
GRANT SELECT ON healthcareprovider TO healthcare_provider_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON healthcareprovider TO admin_role;
-- Granting access to roles for Doctor table
GRANT SELECT ON doctor TO doctor_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON doctor TO admin_role;
-- Granting access to roles for Patient table
GRANT SELECT ON patient TO patient_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON patient TO admin_role;
-- Granting access to roles for Insuranceprovider table
GRANT SELECT ON insuranceprovider TO insurance_provider_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON insuranceprovider TO admin_role;
-- Granting access to roles for Integratedcaredepartment table
GRANT SELECT ON integratedcaredepartment TO healthcare_provider_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON integratedcaredepartment TO admin_role;
-- Granting access to roles for Department table
GRANT SELECT ON department TO healthcare_provider_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON department TO admin_role;
-- Granting access to roles for Laboratory table
GRANT SELECT ON laboratory TO healthcare_provider_role;
```

```
GRANT SELECT, INSERT, UPDATE, DELETE ON laboratory TO admin_role;
-- Granting access to roles for Appointment table
GRANT SELECT, INSERT, UPDATE, DELETE ON appointment TO healthcare_provider_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON appointment TO patient_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON appointment TO admin_role;
-- Granting access to roles for Prescription table
GRANT SELECT, INSERT, UPDATE, DELETE ON prescription TO doctor_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON prescription TO patient_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON prescription TO admin_role;
-- Granting access to roles for Dischargesummary table
GRANT SELECT, INSERT, UPDATE, DELETE ON dischargesummary TO doctor_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON dischargesummary TO patient_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON dischargesummary TO admin_role;
-- Granting access to roles for MedicalRecord table
GRANT SELECT, INSERT, UPDATE, DELETE ON medicalrecord TO doctor_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON medicalrecord TO patient_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON medicalrecord TO admin_role;
-- Granting access to roles for Surgery table
GRANT SELECT, INSERT, UPDATE, DELETE ON surgery TO doctor_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON surgery TO admin_role;
-- Granting access to roles for Billing table
GRANT SELECT, INSERT, UPDATE, DELETE ON billing TO healthcare_provider_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON billing TO patient_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON billing TO admin_role;
-- Granting access to roles for Allergy table
GRANT SELECT, INSERT, UPDATE, DELETE ON allergy TO patient_role;
GRANT SELECT, INSERT, UPDATE, DELETE ON allergy TO admin_role;
------END-----
```

14. Queries

Below are sample SELECT queries to retrieve data from various tables in the Public Health Ontario System database, ensuring that the data is related and valid.

Retrieve Patient Information with Associated Medical Records:

```
SELECT Patients.PatientID, Patients.PatientFName, Patients.PatientLName,
MedicalRecords.Diagnosis, MedicalRecords.DateOfVisit
FROM Patients
JOIN MedicalRecords
ON Patients.PatientID = MedicalRecords.PatientID;
```

Retrieve Appointment Details with Patient and Provider Information:

```
SELECT Appointments.AppointmentID, Appointments.AppDateAndTime,
Patients.PatientFName, Patients.PatientLName, HealthCareProviders.HPLName
FROM Appointments
JOIN Patients ONAppointments.PatientID = Patients.PatientID
JOIN HealthCareProviders ONAppointments.ProviderID =
HealthCareProviders.ProviderID;
```

Retrieve Prescription Details with Patient, Doctor, and Medication Information:

```
SELECT Prescriptions.PrescriptionID, Patients.PatientFName, Patients.PatientLName,
Doctors.DocFName, Doctors.DocLName, Medications.MedicationName
FROM Prescriptions JOINPatients ON Prescriptions.PatientID = Patients.PatientID
JOIN Doctors ONPrescriptions.DoctorID = Doctors.DoctorID
JOIN Medications
ON Prescriptions.MedicationID = Medications.MedicationID;
```

Retrieve Billing Information with Patient and Provider Details:

```
SELECT Billing.BillID, Patients.PatientFName, Patients.PatientLName,
HealthCareProviders.HPLName, Billing.BillAmount
FROM Billing
JOIN Patients ONBilling.PatientID = Patients.PatientID
JOIN HealthCareProviders
ON Billing.ProviderID =HealthCareProviders.ProviderID;
```

Retrieve Laboratory Information with Associated Equipment and Department Details:

```
SELECT Laboratories.LabID, Laboratories.LabName,
MedicalEquipment.EquipmentName, Departments.DepartmentName
FROM Laboratories
JOIN MedicalEquipment ONLaboratories.EquipmentID =
MedicalEquipment.EquipmentID
JOIN Departments ONLaboratories.DepartmentID = Departments.DepartmentID;
```

15. Challenges

While the specifics of challenges in the Public Health Ontario Systems Database project can vary based on the project's unique circumstances, common challenges in database projects generally include:

1. Safeguarding and regulating access to confidential patient information is a primary concern.
2. Optimizing billing procedures and maintaining precise insurance records are essential for efficient operations.
3. Bolstering system scalability and resilience is imperative, ensuring data integrity under various threats, including natural disasters and cyber-attacks.
4. Vigilantly monitoring system performance and implementing timely updates are key for user satisfaction and enhanced productivity.
5. Enforcing robust information security measures is crucial to safeguard sensitive patient data and prevent unauthorized access or breaches.
6. Analyzing system usage and performance allows for proactive identification of areas for improvement, enabling data-driven decisions to optimize both performance and patient care outcomes.
7. Ensuring data accuracy and quality in healthcare is pivotal; continuous efforts in data validation and adherence to quality standards remain ongoing priorities.
8. Scaling the system to accommodate the growing database, including healthcare providers, patients, and data, is vital for sustained efficiency.
9. Overcoming the challenge of interoperability with existing healthcare systems demands meticulous planning and seamless integration efforts to facilitate communication and data exchange across the healthcare ecosystem.

16. Conclusion

In conclusion, the successful implementation of the Public Health Ontario System database represents a significant milestone in the realm of public health management within the province. By harnessing the advanced capabilities of Oracle Database technology, the project has not only streamlined data storage, retrieval, and analysis but has also emerged as a cornerstone in the quest for a more efficient and effective public health infrastructure. The meticulous focus on optimizing database queries and utilizing Oracle SQL commands has enabled the generation of insightful reports, exemplified by the ability to discern the volume of orders placed by individual customers within the preceding year.

This level of precision and granularity in data management is invaluable, equipping Public Health Ontario with the tools needed for evidence-based decision-making, strategic planning, and resource allocation.

The comprehensive database system stands as a testament to the project's commitment to fostering resilience and responsiveness in public health, ensuring that critical information is readily available to support the province's healthcare professionals in their mission to safeguard and enhance the well-being of the population.

As the project concludes, its impact resonates not only in the enhanced efficiency of daily operations but also in the enduring capacity to adapt and address emerging public health challenges with agility and precision.